

1. What is random variable and how does it differ from a deterministic variable?

A. • A random variable takes values based on chance

• Its value is uncertain and depends on probability

• A deterministic variable has a fixed, known value

• It does not involve randomness

2. How is mean (expected value) of random variable defined for:

The mean represents the long-run average value of a random variable

• For a discrete distribution

If X takes values $x_1, x_2, x_3, \dots, x_n$ with probabilities $P(x_1), P(x_2), \dots, P(x_n)$,

$$E(X) = \sum_i x_i P(x_i)$$

For continuous distribution

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

3. How is standard deviation related to variance and why is it often preferred in interpretation?

A) Variance measures how much values spread around the mean

$$\text{variance} = \sigma^2 = E[(x - \mu)^2]$$

Standard deviation is the square root of variance

$$\text{standard deviation} = \sigma = \sqrt{\text{variance}}$$

Standard deviation is preferred as it has the same unit as data and easier to interpret than variance